

AIOPS-02: Anomaly Detection

Series: AIOPS — Dynatrace Intelligence | **Notebook:** 2 of 8 | **Created:** May 2026 |
Last Updated: 05/05/2026

Overview

Anomaly detection in Dynatrace is **Predictive AI in action** — the platform learns what normal looks like and surfaces deviations. Five mechanisms cover the practical detection landscape: static thresholds, auto-adaptive thresholds, seasonal baselines, multi-dimensional baselines, and novelty / forecasting.

Most teams over-rely on static thresholds and miss what adaptive and seasonal detection would catch. This notebook walks the five mechanisms, when to use each, and how to test them against your data using the Davis analyzer MCP tools.

Audience: Platform admin tuning detection; SRE writing custom alerts.

Outcome: A working understanding of which detector to pick for each metric type, and live examples against your tenant.

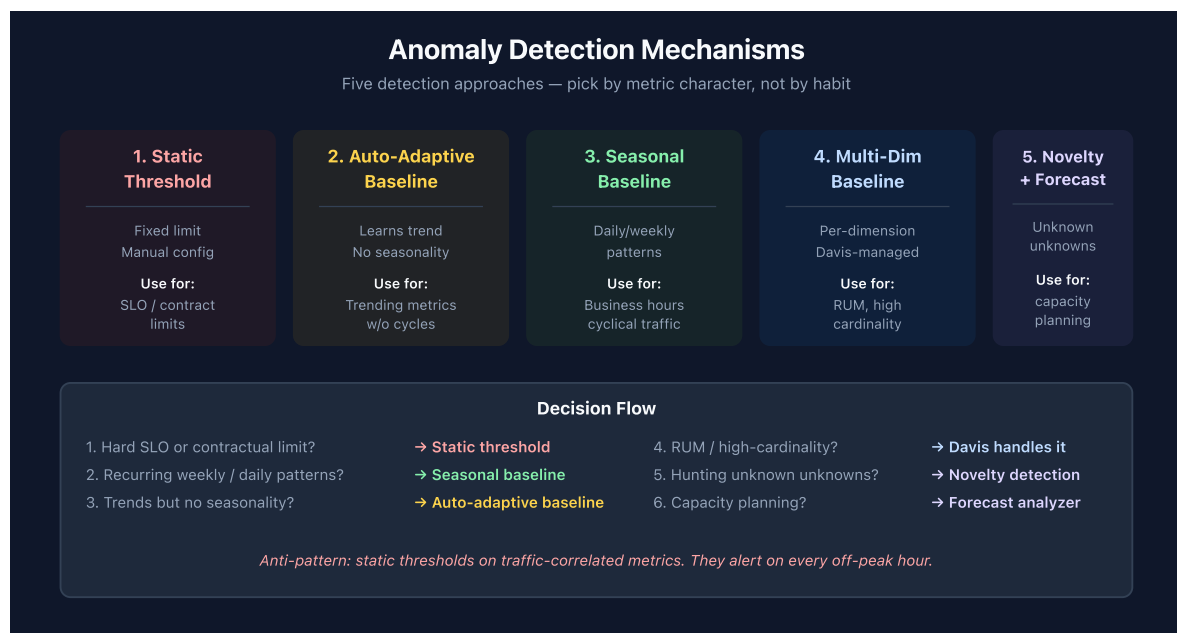


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Prerequisites

Requirement	Details
Dynatrace Environment	SaaS Gen3 with Anomaly Detection app installed
Permissions	<code>davis:analyzers:execute</code> , <code>settings:objects:read/write</code> , <code>events:read</code>
MCP	Dynatrace MCP server (for AIOPS-04 / AIOPS-06 integration); analyzers also exposed in the app
Optional	Monaco / Terraform for config-as-code (see AUTOM-05/06)

1. The Five Detection Mechanisms

1.1 Static threshold

Hard limit: *response time must be < 1 s*. Trips when the metric crosses the line. Configured manually. Fast to set up, but brittle — a threshold that fits Tuesday at 3 AM rarely fits Friday at noon.

Use when: an SLO, contract, or capacity ceiling defines the limit. Don't use for anything that varies with traffic.

1.2 Auto-adaptive threshold

Davis learns the metric's baseline and shifts the detection threshold as the baseline moves. No seasonal awareness — purely adaptive to recent trend.

Use when: the metric trends over time but doesn't have weekly / daily seasonality. Service throughput on a steadily-growing app is a classic fit.

1.3 Seasonal baseline

Davis learns the metric's daily, weekly, and (where it has data) yearly seasonality. Detection considers the *expected* pattern — Tuesday at 3 AM and Friday at noon get different thresholds.

Use when: the metric has obvious recurrence — business hours, weekday/weekend differences, monthly billing cycles.

1.4 Multi-dimensional (automated) baseline

Davis builds baselines per-dimension automatically — per region, per browser, per OS, per user-action. You don't configure this; the platform does it under the hood for RUM-like metrics.

Use when: you don't — Davis chooses. Your job is to let the high-cardinality dimensions through to it (don't pre-aggregate them away).

1.5 Novelty detection and forecasting

Novelty flags patterns the model has never seen — sudden new error types, never-before metric shapes. **Forecasting** projects a series forward, useful for capacity planning and trend extrapolation.

Use when: you're hunting for *unknown unknowns* (novelty) or projecting growth (forecast). Both available as Davis analyzers — see Section 5.

2. Picking the Right Detector

A simple decision flow:

1. *Is there a hard contractual or SLO limit?* → **Static threshold.**

2. Does the metric have recurring weekly/daily patterns? → **Seasonal baseline**.
3. Does it trend over time without strong seasonality? → **Auto-adaptive threshold**.
4. Is it RUM / high-cardinality user-facing? → **Davis handles it; don't pre-aggregate**.
5. Are you hunting unknown unknowns? → **Novelty detection**.
6. Capacity planning? → **Forecasting**.

Anti-pattern alert: static thresholds on traffic-correlated metrics. They alert on every off-peak hour and on every traffic spike. Move to seasonal or auto-adaptive.

3. Configuration Surface: App vs. Settings vs. Code

Three places to configure detection — pick one per environment, not one per detector.

Surface	When to use
Anomaly Detection app	Exploration, one-off detectors, quick wins. Best for SREs in the moment.
Settings 2.0 (UI)	Steady-state detection that's been validated; lives under specific settings schemas.
Config-as-code (Monaco / Terraform)	Anything you want versioned, reviewable, and reproducible across environments.

Production teams should converge on config-as-code. The app is the right place to *discover* what to alert on — but the moment a detector matters in production, it should live in source control. See **AUTOM-05** and **AUTOM-06** for the patterns.

4. Custom Alerts via DQL

Beyond pre-canned detectors, Anomaly Detection app supports **DQL-based custom alerts**. You write the query, the app evaluates it on a schedule, and a breach generates a Davis event.

Example — alert when a service's error rate breaches 5% for 5 minutes:

```
// Custom alert: service error rate > 5% in the last hour
// (When wired into the Anomaly Detection app, this evaluates on a schedule.)
timeseries {
  failures = sum(dt.service.request.failure_count),
  total    = sum(dt.service.request.count)
},
by:{dt.entity.service},
from:-1h, interval:1m
| fieldsAdd error_rate = (failures[] / total[]) * 100
| fieldsAdd avg_error_rate = arrayAvg(error_rate)
| filter avg_error_rate > 5
| sort avg_error_rate desc
| limit 20
```

Watchpoints when writing custom-alert DQL:

- Always include a `from:` time range. Detectors that omit it are rejected.
- Use named-parameter `decimals:`, `then:`, `else:` where required (`round`, `if`, `substring`).
- Aggregations used downstream need explicit aliases (`error_rate = ...`, `mttr = ...`).
- Element-wise array operations on timeseries: `failures[] / total[]` returns an array; wrap in `arrayAvg()` (or similar) to collapse to a scalar before `filter` .

5. Testing Detectors with Davis Analyzers (MCP)

The Dynatrace MCP server exposes Davis analyzers as callable tools. Useful when you want to test a detector against historical data before wiring it to a setting or a workflow.

MCP tool	What it does	Permission
<code>mcp_dynatrace_static-threshold-analyzer</code>	Test a static threshold against a series	<code>davis:analyzers:execute</code>
<code>mcp_dynatrace_seasonal-baseline-anomaly-detector</code>	Detect anomalies vs. seasonal pattern	<code>davis:analyzers:execute</code>
<code>mcp_dynatrace_adaptive-anomaly-detector</code>	Detect anomalies vs. learned baseline	<code>davis:analyzers:execute</code>

MCP tool	What it does	Permission
<code>mcp__dynatrace__timeseries-novelty-detection</code>	Flag never-seen patterns	<code>davis:analyzers:execute</code>
<code>mcp__dynatrace__timeseries-forecast</code>	Forecast future series; capacity planning	<code>davis:analyzers:execute</code>

Workflow pattern: write a `timeseries` query that returns the metric you care about, then pass the query to the analyzer. The analyzer's output is structured (anomalies, forecast bands, novelty flags) and can be embedded in workflows or notebooks.

All five analyzers also have GUI surfaces in the Anomaly Detection app — the MCP tools are the same engine accessed programmatically.

6. Anomaly Volume in Your Tenant

Davis events are the raw signals before Causal AI groups them into problems. Look at the category breakdown to understand what kinds of anomalies your detectors are firing.

```
// Davis event volume by category – last 24h
fetch dt.davis.events, from:-24h
| summarize count = count(), by:{event.category}
| sort count desc
```

```
// Active custom-alert problems (DQL-based custom alerts surface here)
fetch dt.davis.problems, from:-7d
| filter event.category == "CUSTOM_ALERT"
| summarize count = count(), by:{event.status}
| sort count desc
```

Reading the result: A high `CUSTOM_ALERT` count means your custom-DQL detectors are firing — review whether they're noisy. A high `RESOURCE_CONTENTION` or `ERROR` count means out-of-the-box auto-adaptive / seasonal detection is doing real work.

7. Cross-Series Pointers

- **AUTOM-05, AUTOM-06** — anomaly detection settings as code (Monaco, Terraform)
 - **WFLOW** — once a detector fires, route the resulting Davis event into a notification or remediation workflow
 - **DBMON, CLOUD, K8S** — domain-specific anomaly patterns sit in those series; this notebook is the canonical reference for the *mechanisms*
 - **AIOPS-05** — the AI models behind these detectors (causal correlation, predictive baseline, seasonal)
 - **AIOPS-06** — agentic patterns: scheduled analyzer runs in workflows
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This notebook was AI-generated from community-submitted and publicly available sources. This notebook series is not officially supported by Dynatrace. Always verify information against official Dynatrace documentation.